Aurick Anwar

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Education

McMaster University — Hamilton, Ontario
Bachelor of Engineering: Engineering Physics — *Expected 2029*

Work Experience

Software Engineer — Hamilton, Ontario
McMaster Drone Team — *June 2026 – Present*

- Developed an autonomous drone pipeline using **YOLOv11**, **OpenCV**, and **PyTorch**, achieving **30+ FPS** real-time object detection with **90%+ mAP** in **ArduPilot** simulated environments
- Simulated and validated **100+** autonomous flight missions in **NVIDIA Isaac Sim**, testing navigation and obstacle avoidance in different environments before deploying.
- Built **10+ ROS2** nodes, enabling a **<100 ms** communication between perception, planning, and control systems.

Software Engineering Intern — Toronto, Ontario
HermesAI — *May 2026 - Present*

- Built **React** interfaces for document upload and AI-powered case extraction, supporting legal files exceeding **1,000 pages**.
- Developed scalable legal document dashboards, reducing manual processing time by **50% for Canadian law firms.**
- Reduced document processing latency by **30%** by offloading file processing to asynchronous **Node.js.**

Founding Engineer — Toronto, Ontario
Magnified Systems — *February 2026 -Present*

- Built a motorcycle helmet-mounted **IMU (MPU6050)** device to detect and quantify impact severity in real time.
- Trained a **PyTorch** regression model on **1,000+ impact samples**, predicting severity within **±1%**.
- Developed a **C++** severity-score algorithm mapping linear and angular acceleration to a 1–100 impact scale.
- Designed the 3D modelled device enclosure in **SolidWorks**, producing a **PETG** fabricated model.

Personal Projects

↗ **Autonomous Self-Driving Vehicle with CARLA** — *[CARLA, YOLOv11, PyTorch, OpenCV]*

- Developed an end-to-end autonomous driving pipeline in the **CARLA** simulator integrating **YOLO-based object detection** and vehicle control**,** achieving a real-time simulation at **~20-30 FPS**.
- Wrote to a dataset with **5+ features** (steer, brake, speed, acceleration, throttle), generating **25k+** sample datasets to train a **PyTorch** model.
- Used **CrossEntropyLoss**, achieving **~88.5%** prediction directional accuracy for precise automation navigation.

↗ **Breast Cancer Cell Detection** — *[PyTorch, Roboflow, YOLOv11, Grad-Cam, CNNs]*

- Fine-tuned **Roboflow YOLOv11** tech stack on **5,400+** images to localize breast cancer regions with bounding box detection across **benign** and **malignant** classes.
- Built a custom **3-layer PyTorch CNN** classifier achieving **95.31%** confidence on malignant tumor classifications using **CrossEntropyLoss** and **Adam optimization.**
- Implemented **Grad-CAM**, generating **spatial heatmaps** that visually highlight the tumor regions driving each prediction.

Technical Skills

Programming Languages: Python, C++, Javascript, CSS, HTML, MATLAB,
ML and Computer Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, LangChain, Mediapipe
Robotics and Simulation: ROS, RVIZ, Gazebo, SLAM, Nvidia Isaac Sim, ArduPilot
Electronics and Microcontrollers: Arduino, Raspberry Pi, ESP32
3D Design and CAD: Solidworks, Inventor, Fusion360, AutoCAD
Web Development: Node.js, React
Tools and OS: Linux, Git, Docker